



# Software Engineer Intern - Technical Assessment

Assessment Duration: **72 Hours**

## Challenge: Smart Order Allocation System

As part of the selection process for the **Software Engineer Intern** position, you are required to build a small full-stack application based on the scenario below.

The purpose of this assessment is to understand your **technical knowledge, logical thinking, problem-solving approach, coding practices, security awareness, and ability to build a complete working solution.**

## Scenario

A business operates multiple branches and receives customer orders through an online platform.

1. Products and available stock
  2. Location
  3. Current order/workload level
- When a customer places an order, the system should automatically determine the **most suitable branch to fulfil that order.**
  - You are responsible for designing the allocation logic and deciding how the system should behave in different situations.
  - Your solution should consider realistic scenarios such as **stock availability, multiple eligible branches, branch workload, unavailable items, order cancellations, and other edge cases you believe are important.**
  - There is **no predefined allocation algorithm.** We want to understand how you analyse the problem, determine priorities, and design your own solution.

## Core Application Requirements

- Customer order creation
- Product and quantity selection
- Customer/location information
- Multiple branches with stock information
- Automatic branch allocation
- Order status management
- Basic Admin/Dashboard view
- Search or filtering capability
- Persistent database storage
- Backend APIs
- Responsive frontend
- Proper validation and error handling

You are free to decide the **architecture, database structure, API design, data structures, and implementation approach**.

## Authentication & Security

Implement a reasonable authentication and security layer appropriate to your application.

Your implementation should demonstrate understanding of areas such as:

- Secure password hashing
- Authentication using JWT or a suitable session-based approach
- Authentication/token/session expiry
- Protected application routes and backend APIs
- Basic role-based access control
- Input validation
- Secure handling of application secrets using environment variables
- Basic protection for sensitive endpoints, such as rate limiting
- HTTPS when deploying the application

You are **not required to implement every possible security mechanism**.

We are more interested in whether you understand the security decisions you make and apply them appropriately.

For example, a normal user should not be able to gain administrative access simply by modifying frontend data or directly calling a protected API.

Be prepared to explain your **authentication and security approach** during the technical interview.

## AI / ML Challenge - Additional Advantage

As an additional feature, implement a simple solution that can analyse a customer's message or order note and automatically classify it.

**Possible categories:** Payment Issue | Delivery Issue | Refund/Cancellation | Product Inquiry | General Inquiry

### Example

#### Customer Message:

"My payment was deducted, but my order is not showing."

#### Possible Result:

Category: **Payment Issue**

Confidence: **86%**

Your solution should also consider how to handle predictions where the system is **not confident enough**.

You may choose the **AI/ML approach, model, dataset, libraries, or method** that you believe is suitable. The sample dataset is attached separately.

This section is **not mandatory**, but a successful implementation will receive additional consideration.

## Technology

You may use technologies you are comfortable with.

**Examples:** React / Next.js • Node.js • Java • Python • SQL / NoSQL Databases • REST APIs

The specific framework is less important than the **quality of your solution and your understanding of the technologies you use**.

## Submission

Please provide:

- GitHub Repository URL
- Live Application URL, if deployed
- A short README containing:
  - Technologies used
  - Setup instructions
  - Brief explanation of the system architecture
  - Explanation of your branch allocation logic and why you selected that approach
  - Authentication/security approach
  - Any assumptions or limitations
  - AI/ML approach, if implemented

## Evaluation Areas

Your submission will mainly be evaluated based on:

- Logical Thinking & Problem Solving
- Backend Development & API Design
- Database Design & Data Handling
- Data Structures / Algorithmic Thinking
- Frontend Development & API Integration
- Authentication & Security Awareness
- Code Quality & Project Structure
- Validation & Edge-Case Handling
- AI/ML Understanding, if implemented

Candidates selected for the next stage will attend an **online one-to-one technical interview**.

During the interview, we will discuss your implementation, allocation logic, architecture, security decisions, and other software engineering concepts.

## Important

We are **not expecting a production-ready commercial system**.

Focus on building a **working, well-structured and logical solution** that demonstrates how you approach technical problems.

**Do not focus only on the number of features. We are more interested in how you think, design, secure, and implement your solution.**